

MIE245 - Winter 2026

Lab 7 - Red-Black Trees and Skiplists

Deadline: Tuesday, March 10, 11:59pm EST

Submission Instructions

Reminder: You can not claim work that was completed by others or generative AI as your own. Submit a completed **readme** template file outlining what code you contributed and what code was completed by other students, people, tools, and/or leveraged from websites.

Add `_utorid` (with `utorid` replaced by your UTORid in all lower-case characters) to the end of both the completed template file names (i.e., the `.py` and the `readme` files). Please ensure you use your UTORid (i.e., what you use to sign into Quercus and **not** other identifiers such as your student number).

Compress your completed template files to a zip file and upload the zip file to Quercus.

Lab Instructions

In this lab, you will write the *insert* methods for class implementations of a Red-Black Tree and a Skiplist. Additionally, you will write an experiment class to compare run times of your written insert and the provided delete methods for the two data structures.

Complete the template file **MIE245_Lab_7_Template.py** by finishing the following required methods:

- *insert(data: int)*: Insert data in the RBT. Returns nothing.
- *insert(elem: int)*: Insert an element in the Skiplist. Returns nothing.
- *get_insert_times(data_structure: RedBlackTree or Skiplist)*: Given a list of elements stored in the Experiment class (`self.data`) and a data structure, insert the elements into the data structure and record the runtime to insert each element in a list. Return the list of runtimes.
- *get_delete_times(data_structure: RedBlackTree or Skiplist)*: Given a list of elements stored in the Experiment class (`self.data`) and a data structure, delete the elements from the data structure and record the runtime to delete each element in a list. Return the list of runtimes.
- *plot_insert_times(insert_times_RBT: list, insert_times_SL: list)*: Plot the insert times of the two data structures as two lines on a single plot. The x-axis should be the element inserted and the y-axis should be the runtime in seconds. This method sets `self.insert_plot`.
- *plot_delete_times(delete_times_RBT: list, delete_times_SL: list)*: Plot the delete times of the two data structures as two lines on a single plot. The x-axis should be the element inserted and the y-axis should be the runtime in seconds. This method sets `self.delete_plot`.

- For both plots, give them a clear main title, axis titles, a legend, and make sure the x-axis is categorical (Hint: `ax.set_xticklabels(self.data)`)

AutoGrading

Additionally, you have access to an example of the auto-grading script that will be used to test your submission (**MIE245_Lab_7_Autograder.py**). Please follow the steps below to run the auto-grader for the provided test case:

1. Place the auto-grader script in a new folder
2. Inside the newly created folder, create a folder called “submissions”
3. Place your script in the “submissions” folder
4. Run the auto-grader script and open the **grades.csv** file to see if you passed the test case (1 is pass, 0 is fail)

Note: The auto-grading script only contains one of the several test cases we will evaluate against, so even if you pass this test case, you may not have a completely correct solution.

Tip: You can modify the test case to test additional functionality.